

Idea Economic Impact

Expanded research and economic model for Slide 2. The supplied slide states ₹520 crore/year current losses, ₹2,160 crore/year MOIL savings/production uplift and ₹11,000+ crore/year sector-scale impact. [\[filecite\turn1file0\L14-L21\]](#)

1. Solution concept

The proposed idea combines live data, early flagging, reserve mapping, risk flagging, data fusion and environmental signals into a continuous decision pipeline rather than separate manual processes.

The economic thesis is that value comes from several small improvements accumulating across the mining value chain: fewer poor exploration decisions, earlier disruption warnings, improved equipment utilization, better production planning and better control of ore quality.

2. What the ₹520 crore/year figure must represent

The ₹520 crore/year figure is a team estimate in the supplied submission, not a number independently established by the public sources used in this report. It should therefore be decomposed before being shown as a headline fact.

A defensible decomposition is: exploration savings + avoided downtime + recovered production + blending/grade value + equipment-utilization benefit – system operating cost.

3. Build the MOIL value model

Start with MOIL's actual production scale and published exploration activity. The FY25 report gives 107,530 m exploration drilling and 121.97 Mt reserves/resources. [\[cite\turn0search48\]](#)

For production-side value, use a baseline tonne value or contribution margin supplied by MOIL/authorized project data. Do not infer revenue simply from a spot market price.

For downtime avoidance, calculate hours avoided × production rate per hour × realized contribution per tonne, with conservative confidence intervals.

For exploration, calculate drilling metres avoided or re-prioritized × cost per metre, but only after defining how a target is considered successful.

4. Sector-scale economics

The supplied deck claims over ₹11,000 crore/year at full scale. Treat this as a scenario. A national model should define the number of mines covered, average annual production, expected percentage improvement, grade/value assumptions, implementation cost and adoption rate.

A useful sensitivity analysis should show low/base/high scenarios rather than one large number. For example: 1%, 3%, 5%, 10% production-recovery scenarios, with all price and cost assumptions clearly stated.

5. Verified market context

MOIL's FY25 MD&A states that India produced 3.75 Mt of manganese ore in FY25, compared with 3.38 Mt in FY24, an increase of about 11%. It also reports about 6.58 Mt of manganese ore imports in FY25 versus 5.59 Mt in FY24. [\[cite\turn1search24\]](#)

This gives NAKSHATRA-X a stronger economic narrative: the platform is being proposed in a market where domestic production is growing but imports remain substantial.

6. Economic KPI dashboard

Exploration: cost per successful target; metres drilled per successful target; reserve/resource added per metre.

Production: planned vs actual tonnes; shortfall percentage; recovery after intervention.

Risk: alert lead time; false-alert rate; hours of disruption avoided.

Grade: predicted vs assay grade; constraint violations; off-spec tonnes.

Financial: avoided cost, recovered contribution, implementation cost and ROI.

7. Stronger judge message

“We are not claiming that AI automatically creates ₹X crore. We define exactly where value can be created, measure each component, and validate the total against historical mine data.”

Evidence and exact source paths

ID	Source	Exact resource / URL	Use in NAKSHATRA-X
S1	MOIL 63rd Annual Report 2024-25	https://moil.nic.in/userfiles/Annual_Report_2024_25.pdf	Reserves/resources, exploration drilling, production and strategic target.
S2	MOIL MD&A; / Annual Report copy	https://moil.nic.in/userfiles/file/InvRel/Financials/Annual_Report_2024-25.pdf	FY25 manganese production and import context.
S3	MOSDAC — GSMap ISRO Rain	https://www.mosdac.gov.in/gsmmap-isro-rain	Exact rainfall product; hourly, 0.1° and March 2000 onward.
S4	GSI Reports database	https://gsi.gov.in/gsiReport/	Official geological/exploration report search.
S5	GSI Mineral Hunt data portal	https://hackathon.gsi.gov.in/login	Direct geology, exploration and remote-sensing data routes.
S6	GSI geology data	https://nis.bisag-n.gov.in/download/Geology/	Direct geology data directory.
S7	GSI exploration reports/data	https://nis.bisag-n.gov.in/download/Exploration_data_reports/	Direct exploration data/report directory.
S8	GSI remote sensing ASTER	https://nis.bisag-n.gov.in/download/Remotesensing_ASTER/	Direct remote-sensing directory.
S9	GSI/OGD manganese deposit dataset	https://tn.data.gov.in/catalog/location-manganese-ore-deposits-in-dia-and-its-salient-features	Specific manganese-deposit location/geology dataset.
S10	Indian Minerals Yearbook portal	https://ibm.gov.in/IBMPortal/pages/Indian_Minerals_Yearbook	Official IBM source for manganese statistics.
S11	Indian Minerals Yearbook 2024	https://ibm.gov.in/writereaddata/files/177426215469c1178a48453IMYB_2024_EBookFinal.pdf	Producer/location and manganese industry data.
S12	ESA — Sentinel-1 seeing through clouds	https://www.esa.int/ESA_Multimedia/Videos/2014/03/Sentinel-1_seeing_through_clouds	SAR capability through cloud/rain and day/night.
S13	ESA — Introducing Sentinel-1	https://www.esa.int/Applications/Observing_the_Earth/Copernicus/Sentinel-1/Introducing_Sentinel-1	C-band SAR, all-weather/day-night mission context.